Overview of the framework.

The framework features a user-friendly graphical interface through which users can load decision tables, generate the corresponding simplified binary matrix, and compute reducts or constructs based on shortest-length or coarsest-granularity optimization criteria.

The framework's main window is depicted in Figure 1.

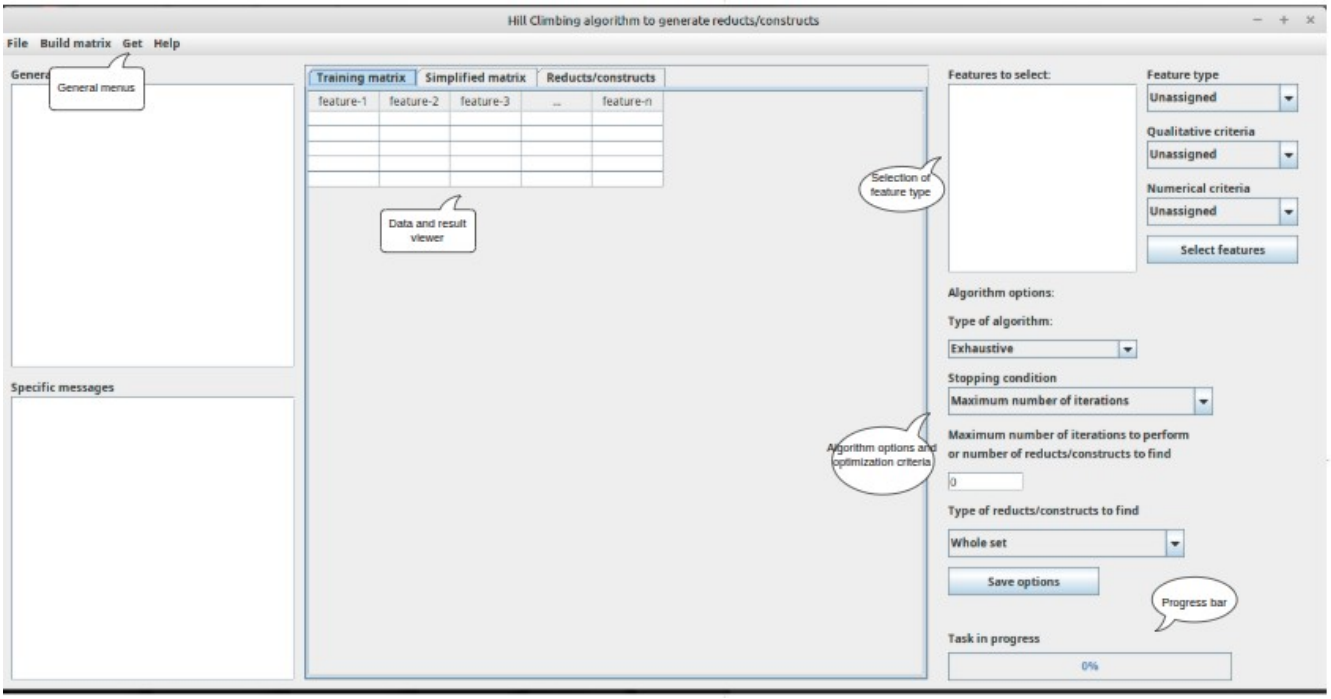


Figure 1. Framework main window.

The main window is divided into five components: (a) the **General Messages** panel, (b) the **Specific Messages** panel, (c) the **Matrix Tabs** area, (d) the **Feature Selection Criterion** panel, and (e) the **Algorithm Options** panel.

The **General Messages** panel displays information related to decision tables loaded from a file and the corresponding simplified binary matrices. It also reports the start and completion of processing tasks. In addition, this panel presents the number of computed reducts or constructs, along with the execution time and the values associated with the shortest-length or coarsest-granularity criteria, depending on the selected options.

The **Specific Messages** panel provides detailed information about the processed decision table, including the number of rows and columns, as well as the selected feature-selection criterion and algorithm.

The **Matrix Tabs** area displays the loaded decision table, the generated simplified binary matrix, and the reducts or constructs computed according to the options selected for the hill-climbing algorithm.

The **Feature Selection Criterion** panel is used to specify the feature types and their corresponding comparison criteria required for generating the simplified binary matrix.

The **Algorithm Options** panel allows the user to define the algorithm's stopping condition and select the type of reduct or construct to be computed.

Once the decision table has been loaded and the feature types and corresponding comparison criteria have been specified, the simplified binary matrix is generated and displayed. This matrix is obtained incrementally from the decision table without explicitly computing the discernibility matrix or the discernibility/similarity matrix. An example of a simplified binary matrix is presented in Figure 2.

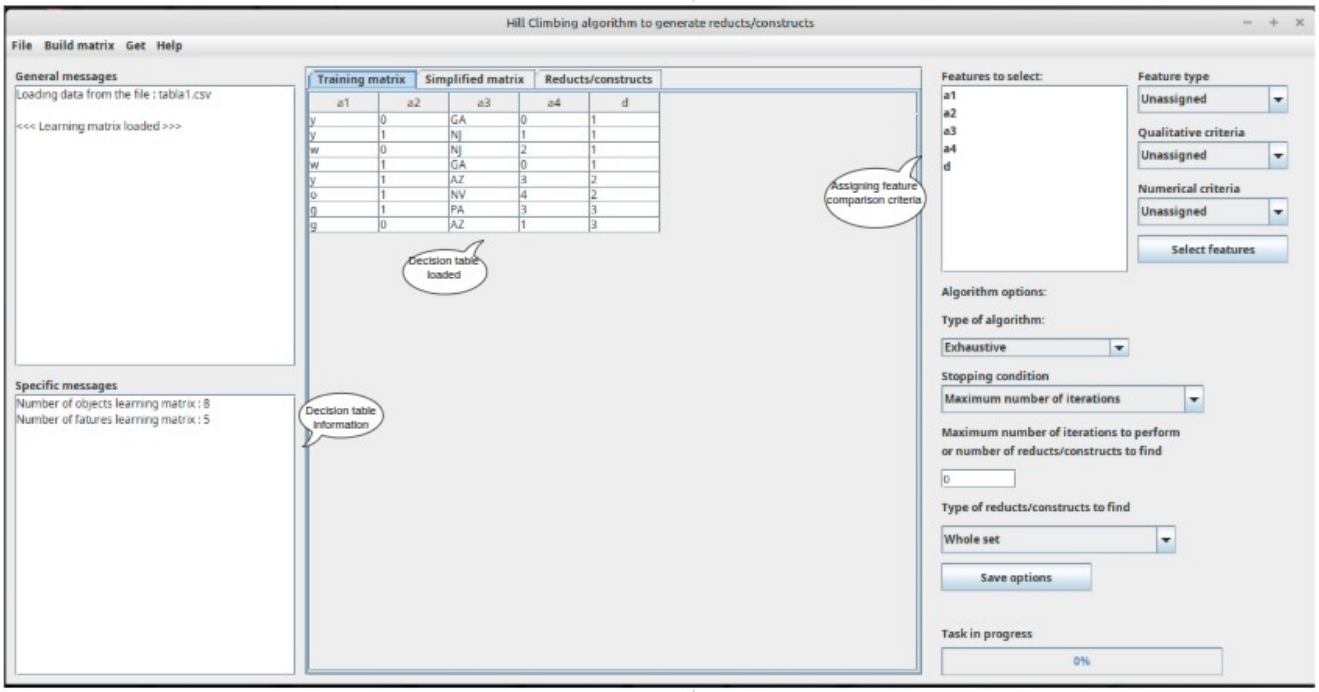


Figure 2. Example of a simplified binary discernibility matrix.

After the algorithm stopping criterion has been selected, the computation of reducts or constructs can be performed. The resulting subsets are displayed in the corresponding interface tab. Figure 3 illustrates an example of the computed reducts.

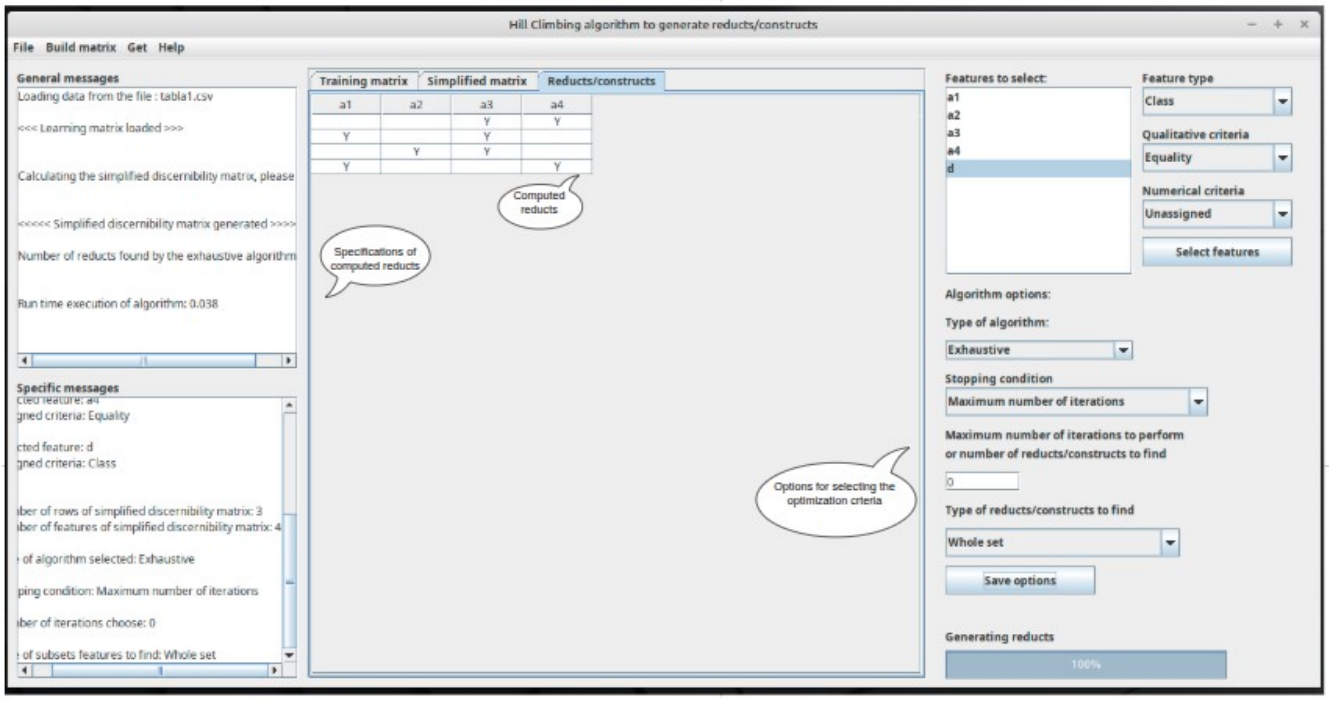


Figure 3. Reducts computed from the simplified binary discernibility matrix.

In addition, both the generated matrix and the computed reducts or constructs can be saved to disk in text format.

The proposed framework guides users through each stage of the process according to the selected options. To ensure the correct execution of the workflow, the framework prevents users from proceeding to subsequent stages until all required options have been specified.

Furthermore, the framework includes a progress bar for both the generation of the simplified binary matrix and the computation of reducts or constructs. This progress bar provides real-time feedback on the advancement of the corresponding task.